are by now many well-written and well-documented code packages available, and it is our firm belief that with a good understanding of the mathematics and the inner workings of the methods, the reader will be able to make the connection between this book and his/her favorite code package in his/her favorite programming language.

We take a statistical perspective in this book, meaning that we discuss and motivate methods in terms of their statistical properties. It therefore requires some previous knowledge in statistics and probability theory, as well as calculus and linear algebra. We hope that reading the book from start to end will give the reader a good starting point for working as a machine learning engineer and/or pursuing further studies within the subject.

The book is written such that it can be read back to back. There are, however, multiple possible paths through the book that are more selective depending on the interest of the reader. Figure 1.6 illustrates the major dependencies between the chapters. In particular, the most fundamental topics are discussed in Chapters 2, 3, and 4, and we do recommend the reader to read those chapters before proceeding to the later chapters that contain technically more advanced topics (Chapters 5–9). Chapter 10 goes beyond the supervised setting of machine learning, and Chapter 11 focuses on some of the more practical aspects of designing a successful machine learning solution and has a less technical nature than the preceding chapters. Finally, Chapter 12 (written by David Sumpter) discusses certain ethical aspects of modern machine learning.

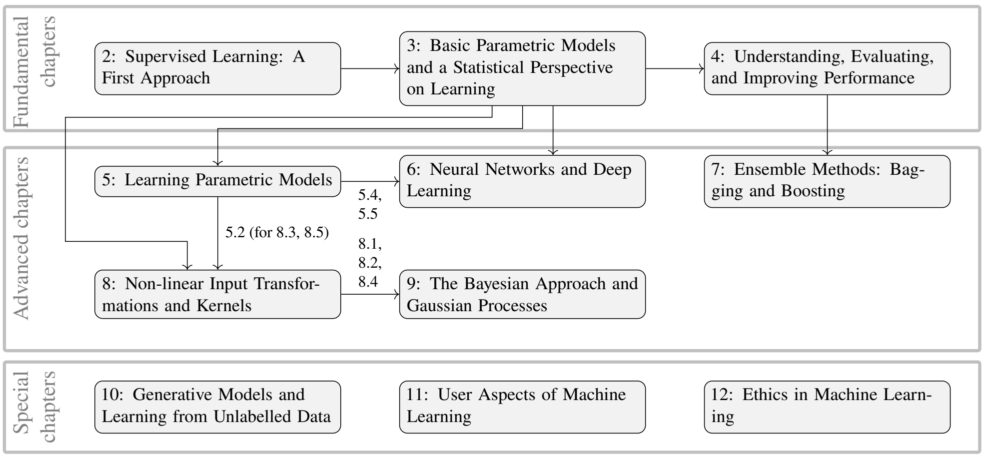

Figure 1.6: The structure of this book, illustrated by blocks (chapters) and arrows (recommended order in which to read the chapters). We do recommend everyone to read (or at least skim) the fundamental material in Chapters 2, 3, and 4 first. The path through the technically more advanced Chapters 5–9, can be chosen to match the particular interest of the reader. For Chapters 11, 10, and 12, we recommend reading the fundamental chapters first.